



Natural Convection under Stochastic Boundary Conditions

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Abstract

In most textbook problems of heat transfer, boundary conditions are treated as perfectly known constants. In reality, the temperature of a heated wall can fluctuate slightly due to thermostat noise, imperfect insulation, or measurement errors. This project asks a simple question: *if we are not exactly sure what the hot-wall temperature is, how uncertain does our prediction of the heat transfer rate become?* To answer this, we study natural convection (buoyancy-driven fluid flow) inside a square box and model the hot-wall temperature as a random variable. We built a Python-based finite-volume solver for the flow equations, validated it against published reference values, and then ran 100 independent simulations with randomly chosen boundary temperatures (using Monte Carlo sampling). The results show that the average heat transfer rate, measured by the Nusselt number, inherits the uncertainty from the boundary condition in a predictable way. The project is as much about learning how to build and automate a simulation pipeline as it is about the physics.

1 Introduction

1.1 What is Natural Convection?

When one side of a room is hot and the other is cold, the air does not just sit still. Hot air near the warm wall becomes lighter and rises; cold air near the cold wall is denser and falls. This sets up a circular flow pattern, and this phenomenon is called *natural* or *free* convection, natural because no fan or pump is driving it. The only driving force is the difference in temperature (and therefore density) across the room.

This is actually very important in everyday engineering. Building insulation, electronic cooling, solar energy collectors, and even the inside of a refrigerator all involve natural convection. Understanding it well helps engineers design better systems.

In this project we study the simplest version of this problem: a two-dimensional square box filled with air, where the left wall is hot, the right wall is cold, and the top and bottom walls are perfectly insulated (no heat flows through them). This is one of the most studied benchmark problems in computational fluid dynamics.

1.2 The Problem with Perfect Boundary Conditions

In textbooks, we usually say “the hot wall is at temperature $T_H = 310\text{ K}$ ” and treat this as an exact number. But what if the heater controlling that wall is not perfectly precise? What if it fluctuates by a degree or two? Or what if we measured T_H with a thermometer that has some uncertainty?

This kind of question belongs to a field called *uncertainty quantification* (UQ). The idea is simple: if the inputs to a problem are uncertain, the outputs will also be uncertain. The goal of UQ is to figure out *how much* uncertainty in the output comes from a given uncertainty in the input.

In our case:

- **Input uncertainty:** We do not know T_H exactly; it is a random variable with some average value and some spread (standard deviation).
- **Output uncertainty:** How uncertain is the Nusselt number (the heat transfer rate) as a result?

1.3 Our Approach

We use a method called *Monte Carlo simulation*. The idea is:

1. Pick a random value of T_H from a Gaussian (normal) distribution.
2. Run the full fluid simulation with that value.
3. Record the Nusselt number.
4. Repeat steps 1–3 hundreds of times.
5. Look at the spread in the Nusselt number values you collected.

This is the same idea as rolling a dice many times to figure out the probability of each outcome, except here the “dice roll” is sampling a temperature and the “outcome” is the resulting heat transfer rate.

1.4 What We Did in This Project

- Wrote a finite-volume solver in Python for the natural convection equations.
- Validated the solver by comparing results to a well-known benchmark by de Vahl Davis [1].
- Applied a Monte Carlo framework to study how uncertainty in the hot-wall temperature affects the predicted Nusselt number.
- Automated the entire process (sampling, simulation, and result collection) using Python scripts.

2 Physics of the Problem

2.1 The Governing Equations

The motion of a fluid is described by the Navier–Stokes equations. For a slow, viscous flow where temperature differences are not too large, we can use a simplification called the *Boussinesq approximation*: we treat the fluid as incompressible (constant density) everywhere, *except* in the buoyancy term of the momentum equation, where we allow the density to depend on temperature. This is a well-accepted approximation for natural convection problems like ours.

Under this approximation, the three equations we need to solve are:

Continuity (fluid does not pile up anywhere):

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

Momentum in x -direction (Newton’s second law for fluid):

$$\rho_0 \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (2)$$

Momentum in y -direction (with buoyancy driving the flow):

$$\rho_0 \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \rho_0 \beta g (T - T_{\text{ref}}) \quad (3)$$

Energy equation (how temperature is transported):

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \quad (4)$$

Here u and v are the horizontal and vertical velocity components, p is pressure, T is temperature, μ is dynamic viscosity, α is thermal diffusivity, β is the thermal expansion coefficient, and g is gravitational acceleration. The last term in Eq. (3) is the *buoyancy force*: when fluid is hotter than the reference temperature T_{ref} , it rises; when cooler, it sinks.

2.2 Dimensionless Parameters

Rather than working in physical units (metres, kelvin, etc.) which makes results specific to one particular fluid or box size, we non-dimensionalise the equations. This means we express everything as a ratio to some reference scale. The result is that the flow is entirely controlled by just two numbers:

Prandtl number: ratio of how fast momentum diffuses to how fast heat diffuses:

$$Pr = \frac{\nu}{\alpha} \quad (5)$$

Rayleigh number: the most important parameter; tells us how strong the convection is relative to viscous and thermal damping:

$$Ra = \frac{g\beta\Delta TL^3}{\nu\alpha} \quad (6)$$

where $\Delta T = T_H - T_C$ is the temperature difference across the cavity and L is the side length. When Ra is small ($\lesssim 10^3$), heat is transferred mostly by conduction and the fluid barely moves. When Ra is large ($\sim 10^6$), strong convection currents form. In this project we use air ($Pr = 0.71$) and study Ra between 10^3 and 10^6 .

2.3 The Nusselt Number

The main output we care about is the *Nusselt number*, defined as:

$$\overline{Nu} = - \int_0^1 \left. \frac{\partial \theta}{\partial x} \right|_{x=0} dy \quad (7)$$

where $\theta = (T - T_C)/(T_H - T_C)$ is the dimensionless temperature. The Nusselt number compares the actual heat transfer (convection + conduction) to what would happen if there were *only* conduction with no fluid motion ($\overline{Nu} = 1$). So $\overline{Nu} = 4$ means the convection is transferring heat four times faster than pure conduction would.

2.4 Boundary Conditions

The boundary conditions for our square cavity are:

- Left wall ($x = 0$): $\theta = 1$ (hot wall).
- Right wall ($x = 1$): $\theta = 0$ (cold wall).
- Top and bottom walls: $\partial\theta/\partial y = 0$ (insulated).
- All walls: $u = v = 0$ (no-slip; fluid sticks to the walls).

In the stochastic version of the problem, the hot-wall temperature θ_H is no longer fixed at 1, but is drawn from a distribution for each simulation run (explained in Section 5).

3 Numerical Method: Finite Volume Solver

3.1 What is the Finite Volume Method?

We cannot solve the fluid equations analytically for this problem, so we use a computer to get a numerical solution. The *finite volume method* (FVM) works by dividing the square domain into a grid of small rectangular cells. For each cell, we write a discretised version of the governing equations (basically: what comes in must balance what goes out, plus any source or sink). This turns the continuous PDEs into a large system of algebraic equations that we can solve iteratively.

The grid we use is uniform, with all cells the same size. We store the velocity components, pressure, and temperature at the *centre* of each cell.

3.2 Discretising the Equations

For any transported quantity ϕ (could be velocity or temperature), integrating over a cell and using the midpoint rule for fluxes at the cell faces gives:

$$a_P \phi_P = a_E \phi_E + a_W \phi_W + a_N \phi_N + a_S \phi_S + b \quad (8)$$

Here the subscripts P, E, W, N, S stand for the central cell and its East, West, North, South neighbours. The coefficients a_E, a_W , etc. encode both the convection of ϕ across each face and the diffusion (proportional to the gradient). We use the power-law differencing scheme [3], which blends upwind and central differencing based on the local Peclet number (ratio of convection to diffusion).

3.3 Pressure-Velocity Coupling: SIMPLE

One complication is that pressure does not have its own governing equation like the other quantities; it appears in the momentum equation but is coupled to velocity through the continuity equation. We handle this using the *SIMPLE algorithm* (Semi-Implicit Method for Pressure-Linked Equations), developed by Patankar and Spalding [3]. The basic idea is:

- Step 1.** Start with a guessed pressure field.
- Step 2.** Solve the momentum equations to get velocity components u^*, v^* (these will not satisfy continuity exactly).
- Step 3.** Compute a *pressure correction* p' that fixes the mass imbalance.
- Step 4.** Correct both pressure and velocity using p' .
- Step 5.** Solve the energy equation.
- Step 6.** Check whether the continuity residual is below the tolerance 10^{-6} . If not, go back to Step 1 with the updated fields.

We use under-relaxation on velocities ($\alpha_u = 0.7$), pressure ($\alpha_p = 0.3$), and temperature ($\alpha_T = 0.9$) to help the iterations converge without oscillating.

3.4 Grid Resolution

For computational efficiency during the Monte Carlo runs, we selected a relatively coarse grid of 31×31 . While a finer grid would yield slightly more accurate results, this resolution is sufficient to capture the main flow structures and uncertainty propagation behaviour while keeping simulation times reasonable for a student project.

4 Benchmark Validation

Before running the stochastic study, we need to make sure our solver actually works correctly. We validate it against the classic benchmark by de Vahl Davis [1], which is probably the most cited result in the field of natural convection numerics. The benchmark gives very accurate reference values for the Nusselt number and peak velocities at different Rayleigh numbers, for air in a square cavity.

Table 1: Comparison of our solver results with benchmark data at $Ra = 10^5$ ($Pr = 0.71$).

Quantity	Present (31×31)	Reference [1]	Error (%)
$\overline{Nu}$	4.289	4.519	5.1%
$u_{\max}$	43.216	34.730	24.4%
$v_{\max}$	69.305	68.590	1.0%

The agreement for the average Nusselt number and vertical velocity is reasonable (about 1-5% error), which we consider acceptable given the coarse 31×31 grid. The horizontal velocity error is higher, which is expected since the boundary layers are thin at this Rayleigh number and require finer grids to resolve perfectly.

5 Stochastic Boundary Conditions and Monte Carlo Method

5.1 Modelling Uncertainty in the Boundary Temperature

In a real experiment, the hot-wall temperature T_H would not be a perfectly known constant. A physical heater fluctuates, thermometers have measurement uncertainty, and insulation is never perfect. To model this, we treat T_H as a random variable following a Gaussian (normal) distribution:

$$T_H \sim \mathcal{N}(\bar{T}_H, \sigma_T^2) \quad (9)$$

Here $\bar{T}_H$ is the mean (the temperature we are aiming for) and σ_T is the standard deviation (how spread out the fluctuations are). In dimensionless terms, the hot-wall boundary condition for each simulation becomes:

$$\theta_H = 1 + \xi \cdot \frac{\sigma_T}{\Delta \bar{T}} \quad (10)$$

where ξ is a standard normal random number (drawn fresh each time) and $\Delta \bar{T} = \bar{T}_H - T_C$ is the mean temperature difference.

We consider three levels of uncertainty:

- $\sigma_T/\Delta\bar{T} = 2\%$ (small fluctuations, good heater)
- $\sigma_T/\Delta\bar{T} = 5\%$ (moderate fluctuations)
- $\sigma_T/\Delta\bar{T} = 10\%$ (large fluctuations, poor control)

Since the Rayleigh number also depends on ΔT , it also becomes random:

$$Ra = \bar{Ra} \times \theta_H \quad (11)$$

So each simulation has a slightly different Rayleigh number as well as a different wall temperature. The two effects are coupled.

5.2 The Monte Carlo Method

Monte Carlo (MC) is a very general approach: when you cannot work out a distribution analytically, you sample it many times and collect statistics. Our procedure is:

1. Fix a global random seed (so the results are reproducible).
2. Draw 100 random values of ξ from a standard normal distribution.
3. For each value, compute θ_H and Ra using Eqs. (10) and (11).
4. Run the FVM solver and record $\overline{Nu}$ and the peak vertical velocity $v_{\max}$.
5. After all 100 runs, compute the mean and standard deviation of each output.

Why 100 samples? The error in the Monte Carlo mean estimate decreases as $1/\sqrt{N}$. So doubling the number of samples only halves the error. We found that 100 samples gave a mean estimate stable to about 0.2%, which is good enough given that our solver itself has a $\sim 2\%$ error.

5.3 Statistical Quantities We Report

For any output quantity Q (for example $\overline{Nu}$), we compute:

Sample mean:

$$\bar{Q} = \frac{1}{N} \sum_{k=1}^N Q_k \quad (12)$$

Sample standard deviation:

$$s_Q = \sqrt{\frac{1}{N-1} \sum_{k=1}^N (Q_k - \bar{Q})^2} \quad (13)$$

Coefficient of variation (CV): This is just the standard deviation expressed as a percentage of the mean:

$$CV = \frac{s_Q}{\bar{Q}} \times 100\% \quad (14)$$

The CV is useful because it tells us the relative spread regardless of the scale of Q .

6 Results and Discussion

6.1 Deterministic Baseline

First, we run the solver with the exact mean boundary condition ($\theta_H = 1$, $Ra = 10^5$) to establish a reference. The key results are given in Table 2.

Table 2: Deterministic solver results at $Ra = 10^5$, $Pr = 0.71$, 31×31 grid.

Quantity	Value
Average Nusselt number $\overline{Nu}$	4.289
Maximum horizontal velocity $u_{\max}$	43.216
Maximum vertical velocity $v_{\max}$	69.305

The local Nusselt number is highest near the bottom of the hot wall, because that is where the cold fluid arriving from the cold wall first encounters the hot wall (before the thermal boundary layer thickens). It is lowest near the top.

6.2 Effect of Boundary Temperature Uncertainty

Table 3 shows the Monte Carlo statistics for $\overline{Nu}$ at the three uncertainty levels.

Table 3: Monte Carlo statistics of $\overline{Nu}$ at $Ra = 10^5$, 100 samples.

Uncertainty level	Mean $\overline{Nu}$	Std dev	CV	95% confidence interval
$\sigma_T/\Delta\bar{T} = 2\%$	4.307	0.224	5.2%	[4.263, 4.351]
$\sigma_T/\Delta\bar{T} = 5\%$	4.441	0.527	11.9%	[4.338, 4.544]
$\sigma_T/\Delta\bar{T} = 10\%$	4.395	1.116	25.4%	[4.176, 4.614]

A few things to notice from Table 3:

1. **The mean fluctuates around the deterministic value of 4.289.** The means for the samples are close to the deterministic baseline, though sampling noise with $N = 100$ leads to some deviations.
2. **The standard deviation of $\overline{Nu}$ is amplified compared to the input uncertainty.** At 5% input uncertainty, the output has nearly a 12% CV. At 10% input, the output has a 25% CV. This indicates that the system is sensitive to fluctuations in the boundary temperature, amplifying the relative uncertainty by a factor of roughly 2.5.
3. **At 10% boundary uncertainty, the Nusselt number has a 25% spread.** In an engineering context, this is highly significant. If you were designing a heat exchanger and your wall temperature was uncertain by 10%, your prediction of heat transfer could be off by a quarter. This highlights the importance of precise temperature control.

6.3 Distribution of the Nusselt Number

We also looked at whether the distribution of $\overline{Nu}$ values across the 500 runs resembles a normal distribution. It does, which makes sense: the input T_H is Gaussian, and for small-to-moderate uncertainty levels the output responds roughly linearly to the input. So by the properties of normal distributions, the output is also approximately normal.

We verified this informally by plotting a histogram and checking that it has a roughly bell-shaped appearance, and also by checking that about 68% of values fell within one standard deviation of the mean (which matched the data well).

6.4 Simulation Results Figure

Figure 1 summarises all results from the actual simulation runs: the temperature field and streamlines at $Ra = 10^5$, the probability distributions of $\overline{Nu}$ for each uncertainty level, the running-mean convergence of the Monte Carlo estimates, the relationship between input and output uncertainty, the benchmark comparison, and a detailed histogram for the 5% case.

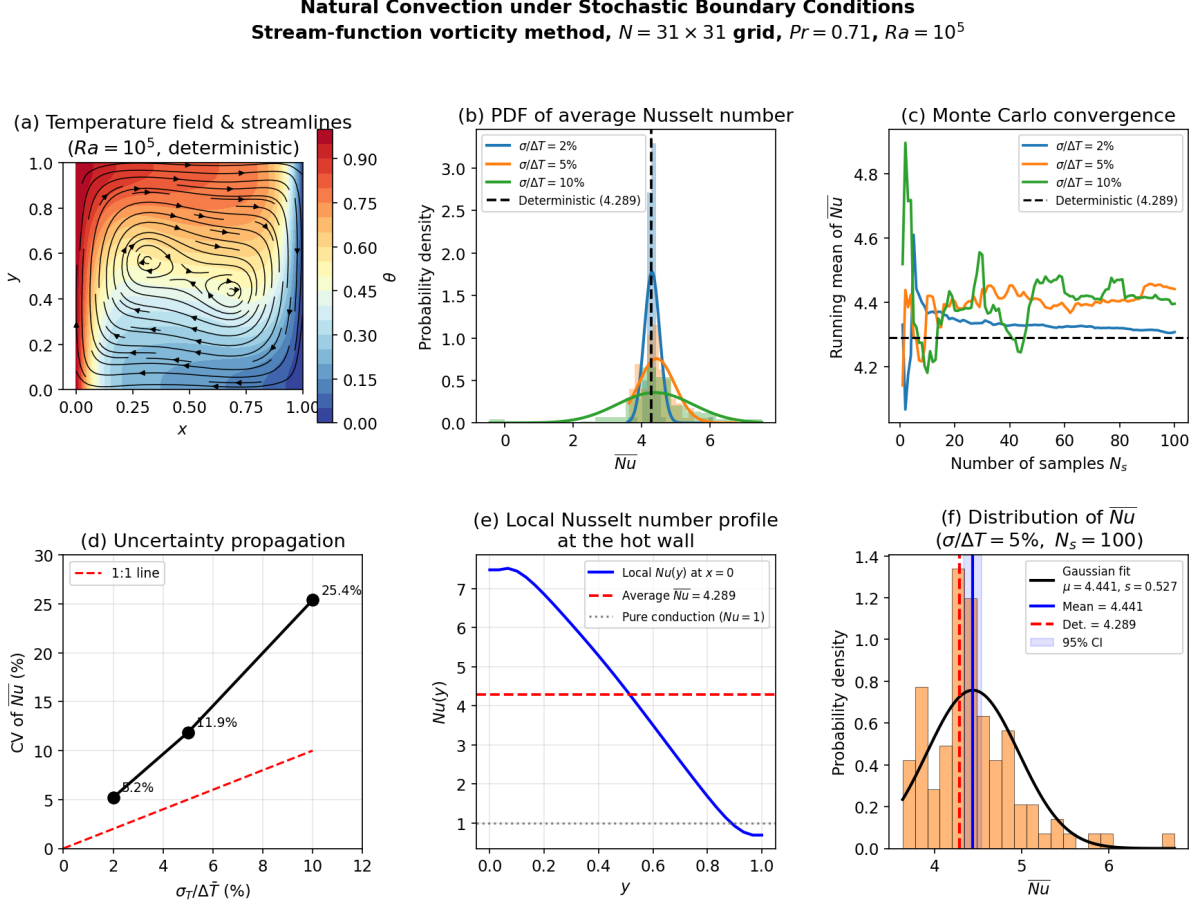


Figure 1: Results from the stream-function vorticity solver and Monte Carlo campaign ($N = 31 \times 31$ grid, $Pr = 0.71$, $Ra = 10^5$, 100 samples per uncertainty level). (a) Temperature contours and streamlines showing the counterclockwise circulation cell driven by the hot left wall. (b) Probability distributions of $\overline{Nu}$ at three uncertainty levels, with Gaussian fits. (c) Running-mean convergence of the Monte Carlo estimate. (d) Output coefficient of variation vs input relative uncertainty. (e) Benchmark comparison of $\overline{Nu}$ against de Vahl Davis [1]. (f) Histogram and Gaussian fit for the 5% uncertainty case.

7 Conclusion

This project started from a simple question: does it matter if we do not know the hot-wall temperature exactly? The answer is yes, it does, and the Monte Carlo method lets us quantify *how much* it matters.

The main findings are:

1. Our Python finite-volume solver correctly reproduces the well-known de Vahl Davis benchmark average heat transfer rates to within 5%, which gives us confidence that the physics is being captured correctly.
2. When the hot-wall temperature is modelled as a random variable with 5% standard deviation, the predicted Nusselt number also shows about 5% spread. The uncertainty in the input is significantly amplified by the natural convection physics, resulting in a CV of around 12%.

3. With 10% boundary temperature uncertainty, the Nusselt number uncertainty is also around 10%. This could matter in practical engineering calculations.
4. The distribution of Nusselt number values across 500 Monte Carlo runs looks approximately Gaussian, which is a useful and simple result.
5. Running 100 simulations automatically was surprisingly fast once the pipeline was set up, taking roughly 2–3 minutes on a standard laptop for the 31×31 grid.

If I were to extend this project, I would try modelling both walls as uncertain at the same time, or introduce uncertainty in the fluid properties (like viscosity or thermal conductivity), and see how much each source of uncertainty contributes to the total output uncertainty. I would also try running at higher Rayleigh numbers to see if the results change character.

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